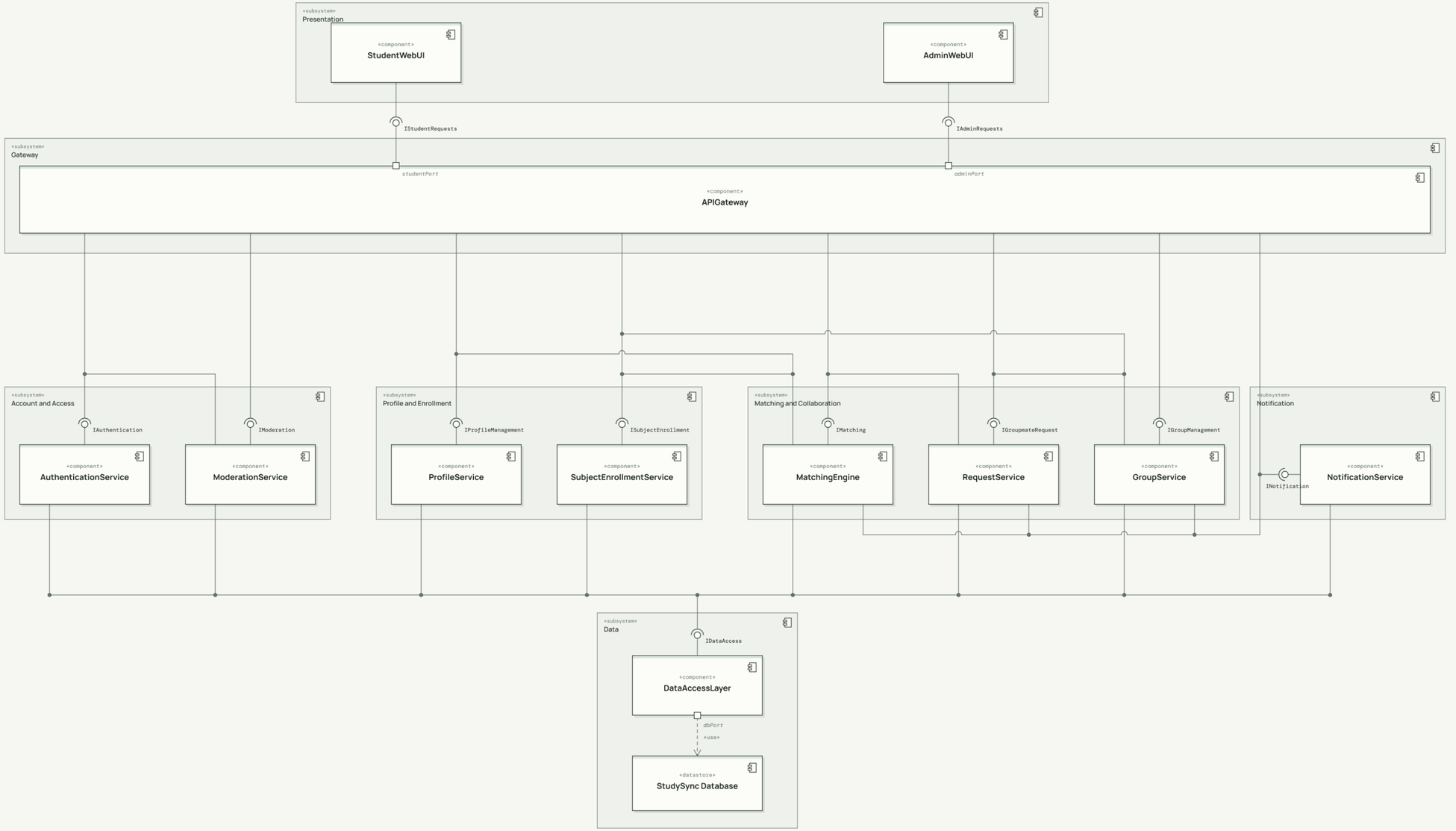
StudySync component diagram

How the system's replaceable parts communicate: every request flows top to bottom from the browser, through the gateway, across the application services, and into the data tier through named interfaces.

COMPONENT · SUBSYSTEM · PROVIDED (BALL) · REQUIRED (SOCKET) · ASSEMBLY CONNECTOR
PORT · SHARED CONNECTOR · NO CONNECTION · «USE» DEPENDENCY



HOW THE COMPONENTS COMMUNICATE

Each path follows assembly connectors in the diagram, from a required socket to the matching provided ball.

1 Sign in

`StudentWebUI → IStudentRequests → APIGateway`
`→ IAuthentication → AuthenticationService`
`→ IDataAccess → DataAccessLayer → StudySync Database`

2 Find potential groupmates

`StudentWebUI → IStudentRequests → APIGateway`
`→ IMatching → MatchingEngine`
`MatchingEngine → IProfileManagement → ProfileService`
`MatchingEngine → ISubjectEnrollment → SubjectEnrollmentService`
`MatchingEngine → INotification → NotificationService`

3 Send a request and form a group

`APIGateway → IGroupmateRequest → RequestService`
`RequestService → IMatching → MatchingEngine`
`APIGateway → IGroupManagement → GroupService`
`GroupService → IGroupmateRequest → RequestService`
`GroupService → ISubjectEnrollment → SubjectEnrollmentService`
`RequestService, GroupService → INotification`

4 Report and moderation

`AdminWebUI → IAdminRequests → APIGateway`
`→ IModeration → ModerationService`
`ModerationService → IAuthentication → AuthenticationService`